

Household's Budget Constraint

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Budget constraint:

Income

- Labor income:

$$p(t) a [1 - u(t)] h$$

price of a service $\nearrow$ $p(t)$

productivity $\nearrow$ a

workers w/ a job $\nearrow$ $[1 - u(t)]$

workers in labor force $\nearrow$ h

services sold by household per unit time

- Investment / saving income:

$$i(t) b(t)$$

nominal interest rate $\nearrow$ $i(t)$

bonds held by household $\nearrow$ $b(t)$

interest from bond holdings

Expenditure

- expenditure on services

$$p(t) [1 + \tau(t)] c(t)$$

price of a service, given by a price menu $\nearrow$ $p(t)$

recruiting wage $\nearrow$ $\tau(t)$

services consumed by household $\nearrow$ $c(t)$

services purchased by household

- Lump-sum tax to finance interest payments

$$T(t)$$

Nominal budget constraint.

$$\dot{\bar{b}}(t) = i(t) \bar{b}(t) + p(t) a [1 - u(t)] \bar{b} - p(t) [1 + \pi(t)] \bar{b} - T(t)$$

change in
savings/nominal
wealth at t .

Budget constraint in real terms:

real stock of bonds. $\underline{w}(t) = \frac{\bar{b}(t)}{p(t)}$

real interest rate $\underline{r}(t) = i(t) - \pi(t)$

nominal
interest
rate

inflation rate = $\frac{\dot{p}(t)}{p(t)}$

$$\frac{d}{dt} \ln(w(t)) = \ln(\dot{\bar{b}}(t)) - \ln(\dot{p}(t))$$
$$\frac{\dot{w}(t)}{w(t)} = \frac{\dot{\bar{b}}(t)}{\bar{b}(t)} - \frac{\dot{p}(t)}{p(t)} = \frac{\dot{\bar{b}}(t)}{\bar{b}(t)} - \pi(t)$$

$$\dot{w}(t) = w(t) \times \left[\frac{\dot{\bar{b}}(t)}{\bar{b}(t)} \right] - \pi(t) \cdot w(t)$$

$$w(t) / \bar{b}(t) = 1 / p(t)$$

$$\dot{w}(t) = \frac{\dot{\bar{b}}(t)}{p(t)} - \pi(t) \cdot w(t)$$

Real budget constraint:

$$\dot{\tilde{w}}_t = \underbrace{r_t w_t}_{\text{interest on wealth}} + a [1 - u_t] h - [1 + \tau(\theta)] c_t - \frac{T_t}{p_t} - \underbrace{T_t w_t}_{\text{tax on wealth}}$$

$$\dot{\tilde{w}}_t = r_t w_t + a [1 - u_t] h - [1 + \tau(\theta_t)] c_t - \frac{T_t}{p_t}$$